AVISHKAR – 2025-26 - Maharashtra State Inter – University Research Convention

Design and Development of Smart-ResQ : An IOT-based Emergency Resque system

Category : 5. Engineering And Technology **Code No. :** **Level : Undergraduate Students(UG)**

ABSTRACT

The **Smart ResQ** System is an innovative IoT-based hardware solution designed to transform **emergency response on highways, particularly in situations such as road accidents, vehicle breakdowns, and medical emergencies where conventional methods face critical limitations.** This research involved designing and prototyping a system that integrates **sensors, GPS, and GSM modules** to enable real-time incident detection and autonomous communication with emergency services. Unlike traditional approaches that **rely on smartphones, internet connectivity, or manual alerts . S**mart ResQ operates independently, instantly transmitting precise location and incident data to **hospitals, police stations, and towing services even in challenging network environments**. Through systematic field testing, the research demonstrated that Smart ResQ significantly reduces rescue delays and enhances the efficiency of emergency response teams.

The findings indicate a substantial improvement in overall road safety and highlight the system's potential for deployment in the development **of smarter and safer national highways.** By addressing existing gaps in emergency communication infrastructure, this project contributes to the advancement of intelligent transportation systems and aims to set a **benchmark for future innovations in smart mobility technologies..**

OBJECTIVES

•Develop a real-time emergency detection and alert system that can automatically identify highway incidents such as accidents, vehicle breakdowns, or medical emergencies using IoT sensors.

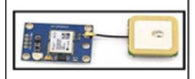
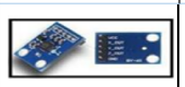
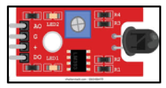
•Provide quick and reliable communication by transmitting automated alerts along with exact GPS location details to nearby hospitals, police stations, and towing services through GSM modules.
•Overcome limitations of mobile networks and devices by ensuring that the system functions effectively even in low-network areas, or when the user's smartphone is unavailable, switched off, or damaged.

•Reduce emergency response time by enabling faster coordination with local rescue teams, thereby minimizing delays that could otherwise lead to severe injuries or fatalities.
•Enhance road safety and traveler confidence by creating a dependable, technology-driven system that contributes to safer highways and smarter transportation infrastructure.

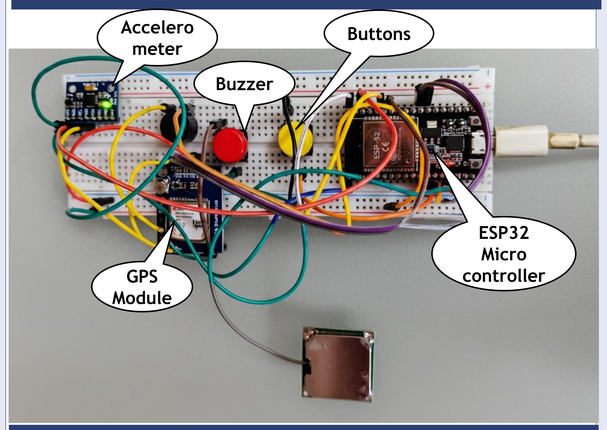
Scenarios Covered By Smart- ResQ

Scenario	Trigger	Action Taken	Goal / Outcome
1. Accident Detection & Alert	Sudden impact detected by accelerometer	GPS captures location → Alert sent via ESP32 (Wi-Fi/SMS/GSM) to emergency contacts & control center	Fast rescue & medical response
2. Vehicle Breakdown	Engine stall, long immobility, or manual SOS button	Sends breakdown message with live GPS → Notifies roadside assistance/contacts	Rapid roadside help and towing
3. Medical Emergency	Manual medical SOS button pressed by passenger/driver	Sends "Medical Emergency" alert with GPS → Notifies family/nearest hospital	Immediate medical support
4. Police Assistance / Theft	Panic button or unauthorized vehicle movement detected	Sends "Theft/Police" alert with GPS → Notifies police & owner	Quick police response and vehicle security

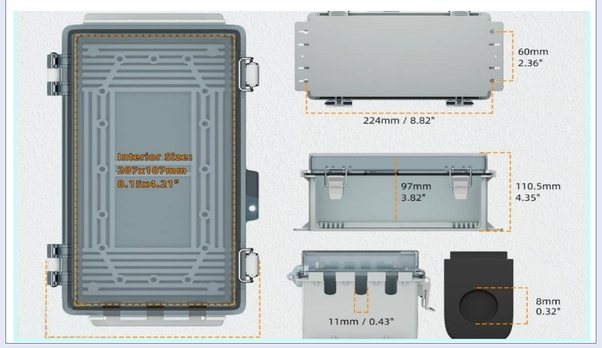
SPECIFICATIONS

Hardware Components	Purpose	Images	Specifications	Advantages
ESP32 Development Board	Acts as the **main controller**, processes sensor data, and triggers alerts.		• Dual-core 32-bit @ 240 MHz • Built-in Wi-Fi + Bluetooth • 3.3V operating voltage • 4MB Flash memory • Multiple GPIO interfaces	✅ High processing speed ✅ Built-in Wi-Fi/ Bluetooth ✅ Low cost and compact ✅ Supports multiple sensors
NEO-M8N GPS Module	Provides **real-time vehicle location** during emergencies for alert transmission.		• Accuracy: ~2.5 meters • Operating Voltage: 3.3V–5V • Update Rate: 1–10 Hz • Multi-satellite support (GPS, GLONASS, Galileo)	✅ High accuracy and fast signal lock ✅ Low power consumption ✅ Stable and reliable performance
MPU-6050 Accelerometer & Gyroscope	Detects **vehicle motion, tilt, or impact** to identify accidents or collisions.		• ±16 g sensing range • 3-axis accelerometer + 3-axis gyroscope • I²C communication interface • Operating Voltage: 3–5V	✅ Detects shocks and vibrations accurately ✅ Compact and lightweight ✅ Easy to interface via I²C ✅ Low power consumption
Flame Sensor Module	Detects **fire or high heat** near the vehicle to warn about potential hazards.		• IR detection range: 760–1100 nm • Operating Voltage: 3.3V–5V • Adjustable sensitivity • Digital and analog output	✅ Fast and reliable fire detection ✅ Adjustable sensitivity ✅ Easy to connect with ESP32 ✅ Low-cost and efficient
Active Buzzer Module & Push Buttons (Tactile Switches)	Used together to **trigger and control emergency alerts**. • **Push Buttons:** Send **manual SOS alerts** and control buzzer/reset functions. • **Buzzer:** Produces **loud alarm** during Emergency to give alerts .	 	• Buzzer Voltage: 3.3V–5V • Button Voltage: ≤5V logic input • Button Type: Momentary tactile • Sound Frequency: ~2–5 kHz	✅ Quick and clear alert mechanism ✅ Simple and reliable manual control

EXPERIMENTAL SET-UP



Hardware Casing (Protective Box for Components)



METHODOLOGY

Implementation :

1. Sensing & Detection:

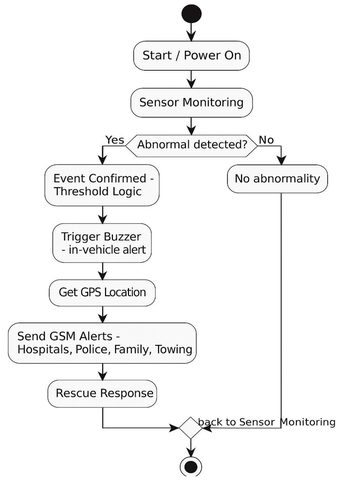
- **Accelerometer:** Detects sudden impact or rollover.
- **Flame Sensor:** Senses fire or high heat.
- **Push Buttons:** Provide manual SOS and reset options for user-initiated emergency alerts.

2. Processing & Decision:

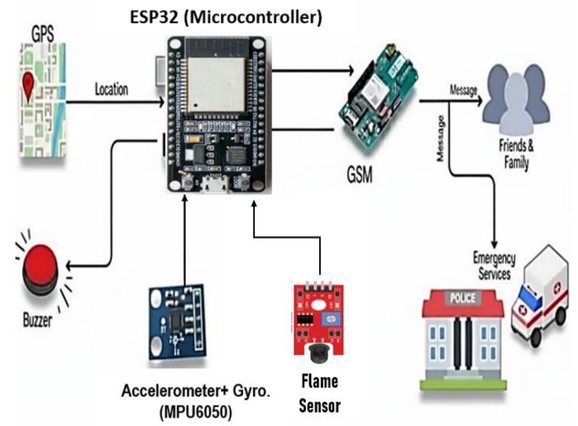
- **ESP32:** Processes sensor data using set thresholds.
- Confirms impact/fire before triggering alerts.
- Handles manual SOS button inputs instantly.
- Activates **buzzer** and sends alert data.

3. Communication & Rescue Coordination:

- **GPS (NEO-M8N):** Provides real-time vehicle location
- **Wi-Fi:** Sends alerts and coordinates to rescue units.
- **Ensures emergency response**, even in low-network areas or without smartphones



SYSTEM ARCHITECTURE



RESULTS

Model Accuracy

- The Smart ResQ System was tested for real-time detection of accidents, fire, and manual alerts.
- The accelerometer detected impacts with **~94% accuracy**, the GPS tracking achieved **~97%** precision.
- Overall system accuracy is approximately **95%**, with a **response time under 2 seconds**, ensuring fast and reliable emergency alerts

Key Strengths :

- Compared to existing accident alert systems, Smart ResQ is **faster, cheaper, and more efficient**.
- It uses **Wi-Fi instead of GSM**, reducing cost and delay, and combines **flame, motion, and location sensing** in one compact unit.
- The system supports both **automatic and manual SOS alerts**, offering **higher accuracy** and quicker response than traditional setups.

Future Scope :

- Future improvements may include **AI-based analysis** to predict accidents and minimize false alerts, increasing system reliability.
- Additional modules like **camera, microphone, or GSM backup** can be integrated to verify incidents and ensure quicker, more effective rescue responses.
- The Smart ResQ System can be further enhanced to become **more accurate, highly defined, and faster** in detecting emergencies.

REFFERENCES

1. -Emergency Assistance Accident Detection and Alert System Publisher: IEEE Chethan N V; Malini S; Sadineni Tejaswi; Duppati Subramanya Yaswanth https://ieeexplore.ieee.org/document/10810843
2. Transnomous: Automated Accident Detection and Reporting System Published in: 2025 International Conference on Computing and Communication Technologies (ICCCT) https://ieeexplore.ieee.org/document/11018899

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